

Based on the **MediCare Connect** repository structure and its integration with [Supabase](#), here is a recommended database schema along with sample seed data tailored to the platform's core workflows (Users, Appointments, Health Records, Prescriptions, and Notifications).

Database Schema (PostgreSQL / Supabase)

SQL

-- 1. ENUMS FOR ROLE & STATUS MANAGEMENT

```
CREATE TYPE user_role AS ENUM ('patient', 'doctor', 'staff', 'admin');
```

```
CREATE TYPE appointment_status AS ENUM ('scheduled', 'completed', 'cancelled', 'in-progress');
```

-- 2. PROFILES TABLE (Extends Supabase Auth / User Metadata)

```
CREATE TABLE profiles (  
  id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,  
  full_name TEXT NOT NULL,  
  role user_role NOT NULL DEFAULT 'patient',  
  email TEXT UNIQUE NOT NULL,  
  phone TEXT,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

-- 3. DOCTORS TABLE (Specialty & Operational Info)

```
CREATE TABLE doctors (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  profile_id UUID REFERENCES profiles(id) ON DELETE CASCADE,  
  specialization TEXT NOT NULL,  
  license_number TEXT UNIQUE NOT NULL,  
  consultation_fee DECIMAL(10, 2) NOT NULL,  
  is_available BOOLEAN DEFAULT TRUE  
);
```

-- 4. APPOINTMENTS TABLE

```
CREATE TABLE appointments (  

```

```
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
patient_id UUID REFERENCES profiles(id) ON DELETE CASCADE,
doctor_id UUID REFERENCES doctors(id) ON DELETE CASCADE,
appointment_date TIMESTAMP WITH TIME ZONE NOT NULL,
status appointment_status DEFAULT 'scheduled',
notes TEXT,
created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

-- 5. ELECTRONIC HEALTH RECORDS (EHR)

```
CREATE TABLE health_records (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
patient_id UUID REFERENCES profiles(id) ON DELETE CASCADE,
doctor_id UUID REFERENCES doctors(id) ON DELETE SET NULL,
diagnosis TEXT NOT NULL,
symptoms TEXT,
record_date TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

-- 6. PRESCRIPTIONS TABLE

```
CREATE TABLE prescriptions (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
appointment_id UUID REFERENCES appointments(id) ON DELETE CASCADE,
patient_id UUID REFERENCES profiles(id) ON DELETE CASCADE,
doctor_id UUID REFERENCES doctors(id) ON DELETE CASCADE,
medication TEXT NOT NULL,
dosage TEXT NOT NULL,
frequency TEXT NOT NULL,
instructions TEXT,
issued_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

-- 7. NOTIFICATIONS TABLE

```
CREATE TABLE notifications (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID REFERENCES profiles(id) ON DELETE CASCADE,  
  title TEXT NOT NULL,  
  message TEXT NOT NULL,  
  is_read BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

Row-Level Security (RLS) Policies

To enforce strict data isolation for medical privacy:

SQL

-- Enable RLS on core tables

```
ALTER TABLE health_records ENABLE ROW LEVEL SECURITY;
```

```
ALTER TABLE prescriptions ENABLE ROW LEVEL SECURITY;
```

-- Patients can only view their own health records

```
CREATE POLICY "Patients view own health records"
```

```
ON health_records FOR SELECT
```

```
USING (auth.uid() = patient_id);
```

-- Doctors can create health records

```
CREATE POLICY "Doctors create health records"
```

```
ON health_records FOR INSERT
```

```
WITH CHECK (
```

```
  EXISTS (
```

```
    SELECT 1 FROM profiles
```

```
    WHERE profiles.id = auth.uid() AND profiles.role = 'doctor'
```

```
  )
```

);

Sample / Project Seed Data

SQL

-- Insert Sample User Profiles

INSERT INTO profiles (id, full_name, role, email, phone) VALUES

('a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11', 'Dr. Sarah Connor', 'doctor', 's.connor@medicare.com', '+15550192834'),

('b1eebc99-9c0b-4ef8-bb6d-6bb9bd380a22', 'John Doe', 'patient', 'john.doe@example.com', '+15550192835'),

('c2eebc99-9c0b-4ef8-bb6d-6bb9bd380a33', 'Alice Smith', 'staff', 'alice.s@medicare.com', '+15550192836');

-- Insert Doctor Details

INSERT INTO doctors (id, profile_id, specialization, license_number, consultation_fee) VALUES

('d3eebc99-9c0b-4ef8-bb6d-6bb9bd380a44', 'a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11', 'Cardiology', 'MD-994821', 120.00);

-- Insert Sample Appointment

INSERT INTO appointments (id, patient_id, doctor_id, appointment_date, status, notes) VALUES

('e4eebc99-9c0b-4ef8-bb6d-6bb9bd380a55', 'b1eebc99-9c0b-4ef8-bb6d-6bb9bd380a22', 'd3eebc99-9c0b-4ef8-bb6d-6bb9bd380a44', NOW() + INTERVAL '2 days', 'scheduled', 'Routine cardiovascular checkup');

-- Insert Sample Health Record

INSERT INTO health_records (patient_id, doctor_id, diagnosis, symptoms) VALUES

('b1eebc99-9c0b-4ef8-bb6d-6bb9bd380a22', 'a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11', 'Mild Hypertension', 'Occasional headaches, elevated blood pressure reading');

-- Insert Sample Prescription

INSERT INTO prescriptions (appointment_id, patient_id, doctor_id, medication, dosage, frequency, instructions) VALUES

('e4eebc99-9c0b-4ef8-bb6d-6bb9bd380a55', 'b1eebc99-9c0b-4ef8-bb6d-6bb9bd380a22', 'a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11', 'Lisinopril', '10mg', 'Once daily', 'Take in the morning with water');

